AEGIS: Equilibrium Structure Mining for Certifiable Graph Robustness

Anonymous Author(s)

***Abstract*—We introduce AEGIS, a framework that mines the equilibrium structure of contractive implicit graph neural networks (IGNN-class models) to analyze adversarial vulnerability under graph structure perturbation. These models converge to a unique fixed point $z^* = F(z^*, A)$, which enables construction of a structural sensitivity matrix $S = (I - J_z)^{-1} J_A$ via the implicit function theorem. We derive a contraction-based vulnerability characterization around a critical perturbation budget $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\|W\|_2$, where $\kappa = \|J_z\|_2 < 1$ is the operator-norm contraction constant. Below this budget, fixed-point shifts are bounded with tight first-order predictions; above it, the contraction guarantee becomes void. The central technical contribution is a constrained sensitivity matrix S_c that restricts analysis to symmetric, edge-only perturbations. Under these realistic constraints, first-order predictions achieve tightness 1.00 ± 0.01 at $\varepsilon = 0.01$ across 9 datasets (5 graph benchmarks and 4 IEEE power grids, 10 seeds each). From S_c, AEGIS derives SVD-optimal structural attacks that inflict 20–50% more damage than adaptive white-box PGD, and deterministic first-order per-node robustness radii that reflect actual vulnerability structure. As a case study, the vulnerability spectrum recovers power grid N-1 contingency rankings with precision@10 between 0.66 and 0.81. Code will be released upon publication.**

***Index Terms*—Deep equilibrium models, graph neural networks, adversarial robustness, structural sensitivity, graph perturbation, implicit function theorem**

I. Introduction

Graph neural networks are vulnerable to adversarial structural perturbation: small changes to graph topology can drastically alter predictions [1], [2], [3], [4]. Defending against such attacks requires understanding which perturbations matter and by how much. Empirical defenses such as robust aggregation [5], [6] and graph structure learning [7], [8] improve resilience but lack formal guarantees. Certified approaches either provide probabilistic bounds via randomized smoothing [9], [10], require layer-by-layer propagation incompatible with infinite-depth models [11], [12], [13], or target explicit architectures exclusively [14], [15].

We observe that contractive implicit GNNs (IGNN-class [16]) possess a structural property that enables a qualitatively different kind of analysis. Rather than propagating through a fixed number of layers, these models iterate a contractive operator F_θ until convergence to a fixed point $z^* = F_\theta(z^*, A)$. This equilibrium structure, combined with the implicit function theorem (IFT), yields the *structural sensitivity matrix*

$$S = (I - J_z)^{-1} J_A, \tag{1}$$

where $J_z = \partial F / \partial z|_{z^*}$ is the state Jacobian and $J_A = \partial F / \partial \mathrm{vec}(A)|_{z^*}$ is the adjacency Jacobian. The matrix S maps structural perturbations directly to equilibrium shifts: $\Delta z^* \approx S \cdot \mathrm{vec}(\delta A)$.

The central finding is that a *constrained* variant S_c of this matrix, restricted to symmetric edge-only perturbations (the physically meaningful threat model for graphs), predicts adversarial vulnerability with tightness 1.00 ± 0.01 to first order at small perturbation budgets ($\varepsilon = 0.01$). This holds across 9 datasets spanning 4 domains and 10 random seeds. At larger budgets ($\varepsilon > 0.1$), higher-order terms dominate and the approximation becomes conservative. We emphasize that AEGIS applies specifically to contractive implicit GNNs; explicit architectures such as GAT, GIN, and GraphSAGE lack the fixed-point structure required for this analysis.

Contributions. We make three contributions:

1) **Contraction-based vulnerability characterization.** For contractive implicit GNNs [16], we formalize vulnerability regimes around a critical budget $\varepsilon_{\mathrm{crit}} = (1 - \kappa)/\|W\|_2$, where $\kappa = \|J_z\|_2 < 1$ is the operator-norm contraction constant. We state assumptions explicitly (ReLU activation, spectral-norm constraint, linear classification head) and quantify non-normality effects (section III).
2) **Constrained structural sensitivity.** We construct the constrained sensitivity matrix S_c that restricts analysis to symmetric, edge-only perturbations (section IV). Under these constraints, first-order predictions achieve tightness ≈ 1.00. From S_c we derive SVD-optimal structural attacks (20–50% stronger than adaptive white-box PGD, section V-D) and deterministic first-order per-node robustness radii (section V).
3) **Cross-domain validation.** We validate across 9 datasets in 4 domains: citation networks [17], e-commerce [18], encyclopedia [19], and IEEE power grids [20]. Constrained first-order tightness ≈ 1.00 holds across all domains. The vulnerability spectrum recovers power grid N-1 contingency rankings with precision@10 = 0.66–0.81 (section VI).

II. Preliminaries and Threat Model

This section establishes the notation, formalizes the model class under consideration, specifies the threat model, and states the problem that AEGIS addresses.

A. Notation and Formalization

Let $G = (V, E, A)$ denote an undirected graph with $N = |V|$ nodes, edge set E, and adjacency matrix $A \in \mathbb{R}^{N \times N}$. Each node $v \in V$ carries a feature vector $x_v \in \mathbb{R}^{d_{\mathrm{in}}}$, collected into $X \in \mathbb{R}^{N \times d_{\text{in}}}$. We write $\hat{A} = D^{-1/2}(A+I)D^{-1/2}$ for the symmetric normalized adjacency with self-loops, where D is the degree matrix of $A+I$.

Deep equilibrium GNNs. A DEQ-GNN [21], [22] defines a weight-tied operator $F_\theta : \mathbb{R}^{N\times d} \times \mathbb{R}^{N\times N} \to \mathbb{R}^{N\times d}$ and iterates it from $Z_0 = 0$ until convergence:

$$Z_{k+1} = F_\theta(Z_k, A), \quad z^* = \lim_{k\to\infty} Z_k. \tag{2}$$

When F_θ is contractive in operator norm ($\kappa = \|J_z\|_2 < 1$, where $J_z = \partial F/\partial z|_{z^*}$), the Banach fixed-point theorem guarantees a unique equilibrium $z^* = F_\theta(z^*, A)$ regardless of initialization.

IGNN instantiation. The Implicit Graph Neural Network (IGNN) [16] uses the operator

$$F(Z, A) = \sigma(\hat{A}ZW^\top + X_{\text{proj}}), \tag{3}$$

where $W \in \mathbb{R}^{d\times d}$ is the shared weight matrix, $X_{\text{proj}} \in \mathbb{R}^{N\times d}$ is a learned feature projection, and σ is ReLU activation. Contractivity is enforced by constraining $\|W\|_2$ via spectral normalization [23], ensuring $\kappa \le \left\|\hat{A}\right\|_2 \cdot \|W\|_2 < 1$. Training uses implicit differentiation through the fixed point [24], [25], and Jacobian regularization [26] can further stabilize convergence. A linear classification head $f(z^*) = W_{\text{cls}}z^* + b$ maps the equilibrium to class predictions.

Implicit function theorem. At the equilibrium, the equation $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A. The IFT gives the sensitivity of z^* to any parameter p:

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p}\bigg|_{z^*}, \tag{4}$$

where the resolvent $(I - J_z)^{-1}$ exists and satisfies $\left\|(I - J_z)^{-1}\right\|_2 \le 1/(1-\kappa)$ by the Neumann series (convergent because $\kappa < 1$). This bound uses the operator norm $\kappa = \|J_z\|_2$; the weaker spectral-radius bound $1/(1-\rho)$ holds only for normal J_z. The pseudospectral index $\eta = \left\|(I - J_z)^{-1}\right\|_2 \cdot (1-\rho) \ge 1$ [27] quantifies this gap, with $\eta = 1$ for normal matrices. The IFT has been used for hyperparameter optimization [28] and implicit differentiation [24], but not for adversarial structural analysis.

B. Threat Model

We consider a white-box attacker who has full access to the trained IGNN model and perturbs the normalized adjacency matrix:

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \text{subject to} \quad \left\|\delta\hat{A}\right\|_F \le \varepsilon. \tag{5}$$

The perturbation is further constrained to be:

- **Symmetric:** $\delta\hat{A} = \delta\hat{A}^\top$ (preserving the undirected nature of the graph).
- **Edge-only:** $[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$ (only existing edge weights may be modified; no new edges are introduced).
- **Continuous:** edge weights are perturbed continuously in $\mathbb{R}$, not discretely flipped.

This threat model captures continuous edge-weight perturbations to the normalized message-passing operator. It aligns naturally with the IGNN architecture, where $\hat{A}$ appears directly in F (eq. (3)). Discrete edge insertions or deletions, which change the graph topology and require recomputing the degree normalization $D^{-1/2}$, are outside the formal guarantee.

The key distinction from input-feature perturbation (studied for DEQs in [29], [30]) is that structural perturbation changes the *operator itself*: the equilibrium depends on A through both the state Jacobian $J_z = \text{diag}(\sigma') \cdot (\hat{A} \otimes W)$ and the adjacency Jacobian $J_A = \partial F/\partial \text{vec}(A)$. This dual dependence makes structural sensitivity qualitatively different from input sensitivity.

C. Problem Statement

Given a trained contractive implicit GNN with equilibrium z^* on graph G, we aim to answer three questions:

1) **Vulnerability ranking:** Which edges, if perturbed, cause the largest shift in the equilibrium? That is, find a ranking of edges $(i, j) \in E$ by their adversarial impact on z^*.
2) **Optimal attack:** What is the worst-case perturbation $\delta\hat{A}^*$ that maximizes $\|\Delta z^*\|_F$ subject to the threat model constraints in eq. (5)?
3) **Robustness assessment:** For each node v, what is the largest perturbation budget r_v such that any $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of v?

AEGIS addresses all three questions through a unified object: the constrained sensitivity matrix S_c derived from the IFT applied to the equilibrium equation. The following sections formalize this construction (section III) and describe the computational pipeline (section IV).

III. Adversarial Equilibrium Theory

Having formalized the problem in section II, we now derive the theoretical foundation for answering the three questions posed there. The main result (theorem 1) characterizes vulnerability regimes as a function of perturbation magnitude, while the constrained sensitivity matrix S_c provides the unified object from which vulnerability rankings, optimal attacks, and robustness radii are all extracted.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, A) = \sigma(AZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

(A1) *σ is ReLU (or any 1-Lipschitz activation with $\sup|\sigma'| = 1$);*
(A2) *W is spectral-norm constrained: $\|W\|_2 \le c$;*
(A3) *The state Jacobian satisfies $\|J_z\|_2 \le \kappa < 1$ at the fixed point z^* (contraction in operator norm).*

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{6}$$

Then structural perturbations δA with $\|\delta A\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical *($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{7}$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}} / (1 - \kappa)$.

(b) Critical *($\varepsilon \to \varepsilon_{\text{crit}}$): The resolvent norm $\left\|(I - J'_z)^{-1}\right\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$. Intuitively, as the perturbation approaches the critical budget, the system's sensitivity to further perturbation grows without bound.*

(c) Supercritical *($\varepsilon > \varepsilon_{\text{crit}}$): The sufficient contraction certificate becomes void ($\|J'_z\|_2$ may exceed 1). The perturbed operator may still be contractive depending on perturbation direction, but the Banach fixed-point theorem no longer guarantees uniqueness, and the first-order guarantees from part (a) cease to apply.*

The intuition behind this three-regime picture is physical: a contractive system has a restoring force that pulls trajectories toward the equilibrium. Small perturbations shift the equilibrium smoothly (subcritical). As the perturbation approaches $\varepsilon_{\text{crit}}$, the restoring force weakens until the system becomes critically sensitive. Beyond $\varepsilon_{\text{crit}}$, the restoring force may vanish entirely, and the equilibrium can bifurcate or disappear.

Proof. **Part (a).** The Jacobian of F with respect to $\text{vec}(Z)$ takes the form $J_z = \text{diag}(\sigma') \cdot (\hat{A} \otimes W)$, where $\sigma' \in \{0, 1\}^{Nd}$ for ReLU (A1). The operator norm satisfies $\kappa = \|J_z\|_2 \leq \left\|\hat{A}\right\|_2 \cdot \|W\|_2 \cdot \sup |\sigma'| = \left\|\hat{A}\right\|_2 \cdot \|W\|_2$. Applying the IFT to $G(z, A) = z - F(z, A) = 0$ at (z^*, A):

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \text{vec}(\delta A) + O(\|\delta A\|^2). \tag{8}$$

Taking operator norms and using the Neumann series bound $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1 - \kappa)$ (convergent because $\kappa < 1$) yields (7).

For contractivity preservation: the perturbed Jacobian satisfies $\|J'_z\|_2 \leq (\|A\|_2 + \|\delta A\|_F) \cdot \|W\|_2$ (using $\|\cdot\|_2 \leq \|\cdot\|_F$ for the perturbation). The condition $\varepsilon < (1 - \kappa) / \|W\|_2$ ensures $\|J'_z\|_2 < 1$.

Part (b). As $\varepsilon \to \varepsilon_{\text{crit}}$, we have $\|J'_z\|_2 \to 1$, so $\left\|(I - J'_z)^{-1}\right\|_2 \geq 1/(1 - \|J'_z\|_2) \to \infty$. Unlike the spectral-radius bound $1/(1-\rho)$, which requires normality, the operator-norm bound holds for all J_z with $\kappa < 1$. In our experiments, the pseudospectral index η ranges from 1.02 to 1.28 (section V-H), confirming mild non-normality.

Part (c). When $\|J'_z\|_2 \geq 1$, the contraction mapping condition fails. The Banach theorem no longer guarantees existence or uniqueness; the iteration may converge to a different equilibrium, oscillate, or diverge. □

Remark (conservativeness). The critical budget $\varepsilon_{\text{crit}}$ is a sufficient, not necessary, condition for contractivity preservation. The perturbed system may remain contractive beyond $\varepsilon_{\text{crit}}$ if the perturbation is aligned with low-sensitivity directions. The unconstrained bound also uses worst-case $\|\cdot\|_F$ relaxation. Under constrained perturbations, the first-order prediction achieves tightness ≈ 1.00 (section V-A). We use $\kappa = \|J_z\|_2$ rather than the spectral radius $\rho(J_z)$ because the Neumann series requires operator-norm contractivity [31]; for normal J_z the two coincide, while for non-normal J_z the κ-based bound is more conservative but requires no normality assumption.

Proposition 2 (Optimal First-Order Structural Attack)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S. The maximum first-order shift is $\varepsilon \cdot \sigma_1(S)$.*

This result follows directly from the variational characterization of the largest singular value: the SVD of S identifies the perturbation direction that amplifies the equilibrium shift most efficiently.

Threat model and constrained sensitivity. As discussed in section II, realistic graph perturbations are symmetric and edge-only. We perturb the normalized adjacency $\hat{A}$ directly, subject to $\left\|\delta\hat{A}\right\|_F \leq \varepsilon$, symmetry ($\delta\hat{A} = \delta\hat{A}^\top$), and support restricted to existing edges. Discrete edge additions that change topology are outside the formal guarantee.

To enforce these constraints, we construct the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{D \times |E|}$ with column

$$[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i} \tag{9}$$

for each edge $(i, j) \in E$. This reduces the perturbation space from N^2 dimensions to $|E|$ dimensions and enforces symmetry by construction. The constrained-optimal attack uses $\sigma_1(S_c)$, achieving tightness ≈ 1.00 empirically (section V-A). The construction of S_c is the central technical contribution of this work: it transforms unconstrained sensitivity analysis into a tight, practical tool for realistic graph perturbations.

Proposition 3 (Per-Node First-Order Robust Radius)**.** *Assume a linear classification head $f(z) = Wz + b$ (as in standard IGNN). For node v with margin $m_v = f_{y_v}(z^*_v) - \max_{c \neq y_v} f_c(z^*_v) > 0$, the first-order robust radius is*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{10}$$

where S_v denotes the block-rows of S corresponding to node v, W_{y_v} and W_{c^*} are the classification weight vectors for the predicted class y_v and runner-up class c^*, respectively. Any perturbation with $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of node v to first order.

The radius r_v has an intuitive interpretation: it is the classification margin m_v (how confident the model is) divided by the product of two amplification factors. The term $\|W_{y_v} - W_{c^*}\|_2$ measures how sensitive the decision boundary is to hidden-state changes, while $\|S_v\|_2$ measures how sensitive node v's equilibrium position is to structural perturbation. Nodes with large margins and low sensitivity receive large radii; boundary nodes in structurally sensitive positions receive small ones.

These radii are deterministic (not probabilistic), unlike randomized smoothing [9] which holds with probability $1 - \alpha$.

Stage 1: Subgraph Extraction & Equilibrium
↓ BFS ego-subgraph (N nodes) → iterate F to fixed point z^*
Stage 2: Jacobian Computation
↓ Row-by-row backward passes → $J_z \in \mathbb{R}^{D\times D}$, $J_A \in \mathbb{R}^{D\times N^2}$
Stage 3: Sensitivity Construction
↓ Linear solve $(I - J_z)^{-1} J_A \to S$; edge projection → S_c
Stage 4: Vulnerability Analysis
SVD of S_c → attack directions, vulnerability spectrum, radii r_v

Fig. 1: The AEGIS pipeline. Each stage feeds into the next; the bottleneck is Stage 2 (Jacobian computation), which dominates wall-clock time for $N > 50$.

They are first-order guarantees: locally tight for small ε but increasingly conservative as perturbation magnitude grows. Without bounding the second-order remainder $O(\|\delta A\|_F^2)$, they should be interpreted as local sensitivity radii rather than global robustness certificates. section V-D verifies empirically that the radii hold (0% breach rate at $\varepsilon = 0.01$, $< 1\%$ at $\varepsilon = 0.10$).

IV. AEGIS Construction

The theoretical results of section III provide the mathematical foundation for vulnerability analysis, but translating them into a practical tool requires careful engineering. This section describes the AEGIS pipeline: how its components fit together, what each one computes, and how the system scales to real graphs.

A. Architecture Overview

AEGIS takes as input a trained contractive implicit GNN (any IGNN-class model satisfying assumptions A1–A3) together with a target graph and produces three outputs: (i) a per-edge vulnerability spectrum ranking edges by adversarial impact, (ii) SVD-optimal attack perturbations, and (iii) per-node first-order robustness radii. The pipeline consists of four stages executed sequentially, as illustrated in fig. 1.

B. Stage 1: Subgraph Extraction and Equilibrium

Full-graph analysis is infeasible for large graphs because the Jacobian J_z has dimension $D \times D$ where $D = N \cdot d$ (number of nodes times hidden dimension). AEGIS addresses this by extracting a local BFS ego-subgraph of N nodes centered on the target node. The model, trained on the full graph, is then run on this subgraph until convergence to the fixed point $z^* = F(z^*, A_{\text{sub}})$. Convergence is verified by checking the relative residual $\|Z_{k+1} - Z_k\|_F \,/\, \max(\|Z_k\|_F\,, 1) < 10^{-5}$. This localization is justified by the locality of per-node vulnerability: a node's robustness radius depends on its own classification margin and the sensitivity of its immediate neighborhood, both of which are well-captured within a 2–3 hop BFS subgraph (section V-F).

C. Stage 2: Jacobian Computation

The sensitivity matrix $S = (I - J_z)^{-1} J_A$ requires two Jacobian matrices:

State Jacobian $J_z = \partial F / \partial \mathrm{vec}(Z)|_{z^*} \in \mathbb{R}^{D\times D}$. Each row is obtained by a single backward pass through the operator F with respect to the flattened state vector. For IGNN with ReLU activation, this takes the form $J_z = \mathrm{diag}(\sigma') \cdot (\hat{A} \otimes W)$, where $\sigma' \in \{0, 1\}^D$ records the activation pattern at z^*.

Adjacency Jacobian $J_A = \partial F / \partial \mathrm{vec}(A)|_{z^*} \in \mathbb{R}^{D\times N^2}$. This captures how each entry of the adjacency matrix influences the operator output. It is computed analogously via D backward passes, each probing one output dimension.

Both Jacobians are computed using automatic differentiation [32]. The total cost is $O(D)$ backward passes, each of cost $O(D)$, giving $O(D^2)$ overall for this stage.

D. Stage 3: Sensitivity Matrix Construction

Given J_z and J_A, AEGIS constructs the full sensitivity matrix by solving the linear system

$$(I - J_z)\, S = J_A. \tag{11}$$

Since $\kappa = \|J_z\|_2 < 1$ by assumption A3, the matrix $(I - J_z)$ is invertible and well-conditioned with condition number at most $1/(1 - \kappa)$. AEGIS solves this system using a dense LU factorization, which costs $O(D^3)$ but is fast in practice for $D \leq 12{,}800$ (i.e., $N \leq 200$ with $d = 64$).

The full sensitivity matrix $S \in \mathbb{R}^{D\times N^2}$ maps arbitrary vectorized adjacency perturbations to fixed-point shifts. However, realistic graph perturbations are constrained: they must be symmetric (undirected graphs) and supported only on existing edges. AEGIS therefore projects S into the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{D\times|E|}$, whose k-th column (for edge $(i, j) \in E$) is

$$[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}. \tag{12}$$

The summation enforces symmetry: perturbing edge (i, j) affects both A_{ij} and A_{ji} simultaneously. This projection reduces the perturbation space from N^2 dimensions to $|E|$ dimensions and is what enables the tight first-order predictions reported in section V-A.

E. Stage 4: Vulnerability Analysis

From S_c, AEGIS extracts three complementary outputs:

Optimal attack direction. The SVD of $S_c = U\Sigma V^\top$ yields the first-order optimal perturbation as $\delta A^* = \varepsilon \cdot v_1$ (the leading right singular vector of S_c, reshaped to edge weights). This perturbation maximizes the fixed-point shift $\|\Delta z^*\|_F$ subject to the Frobenius-norm budget ε, and achieves a shift of $\varepsilon \cdot \sigma_1(S_c)$.

Per-edge vulnerability spectrum. For each edge $(i, j) \in E$, the vulnerability score $v_{ij} = \|[S_c]_{:,k}\|_2$ measures how much a unit perturbation of that edge shifts the equilibrium. Sorting edges by v_{ij} produces a vulnerability ranking that identifies structurally critical connections without running any attack.

Per-node first-order radii. For node v with classification margin m_v (gap between the predicted and runner-up class logits), the radius from theorem 3 is

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \quad (13)$$

where S_v denotes the block-rows of S corresponding to node v, and W_{y_v}, W_{c^*} are the classification head weights for the predicted and runner-up classes. Any perturbation with $\left\|\delta \hat{A}\right\|_F < r_v$ preserves the classification of node v to first order.

F. Computational Cost

The dominant cost is Stage 2 (Jacobian computation) at $O(D \cdot N^2)$ for J_A, followed by Stage 3 (linear solve) at $O(D^3)$. For the default subgraph size $N = 50$ with $d = 64$ (giving $D = 3{,}200$), the full pipeline completes in 1.3 seconds on a single GPU. The practical limit is $N \approx 200$ ($D = 12{,}800$, 8.1 seconds); see section V-E for detailed timings.

V. Experiments

We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (seeds: 42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999). All experiments are implemented in PyTorch [32]. Graph benchmark experiments use IGNN [16] with hidden dimension $d = 64$, spectral-norm constrained weights [23], and early stopping on validation accuracy over 200 epochs. Power flow experiments use ContractiveGCN-PF, an IGNN variant for AC power flow (section VI).

Datasets. We use five graph benchmarks: Cora (2,708 nodes), Citeseer (3,327), and Pubmed (19,717) from the Planetoid collection [17]; Amazon Photo (7,650) from the Amazon co-purchase graphs [18]; and WikiCS (11,701) from Wikipedia-based benchmarks [19]. For power flow, we use four IEEE standard test cases [20]: case14 (14 buses), case30 (30), case57 (57), and case118 (118), with training data generated via PandaPower [33].

Notation. All formal bounds (theorem 1) use the operator-norm contraction constant $\kappa = \|J_z\|_2$. Tables report the spectral radius $\rho(J_z)$ as a diagnostic quantity. Since non-normality is mild ($\eta = 1.02$–1.28, section V-H), we have $\kappa \approx \rho$ in practice; the $\varepsilon_{\text{crit}}$ values computed with ρ are at most 28% optimistic relative to the κ-based formal threshold.

A. Cross-Domain Adversarial Analysis

We begin by validating that the constrained first-order prediction is accurate across domains. table I reports the tightness ratio (actual equilibrium shift divided by predicted shift at $\varepsilon = 0.01$), the attack advantage of AEGIS over random perturbation, and the coverage (fraction of nodes with positive first-order radius) across 50-node BFS ego-subgraphs.

Tightness is 1.00 ± 0.01 across all datasets, confirming that the constrained first-order prediction is essentially exact under symmetric edge perturbations at $\varepsilon = 0.01$. The attack advantage (3.2–4.1× over random) demonstrates that S_c successfully identifies the most vulnerable perturbation directions.

TABLE I: Cross-domain adversarial analysis (mean ± std, 10 seeds). Constrained tightness ≈ 1.00 confirms first-order accuracy.

Dataset	ρ	Tight.	AtkAdv	$\varepsilon_{\text{crit}}$	Cert%
Cora	$.33_{\pm.14}$	$1.01_{\pm.01}$	$3.6_{\pm.6}$x	$.66_{\pm.15}$	$81_{\pm\ 9}$%
Citeseer	$.59_{\pm.02}$	$1.01_{\pm.01}$	$4.1_{\pm.5}$x	$.41_{\pm.02}$	$83_{\pm\ 4}$%
Pubmed	$.41_{\pm.11}$	$1.01_{\pm.01}$	$3.2_{\pm.4}$x	$.59_{\pm.11}$	$74_{\pm23}$%
Amazon	$.14_{\pm.02}$	$1.00_{\pm.00}$	$3.8_{\pm.2}$x	$.86_{\pm.02}$	$92_{\pm\ 4}$%
WikiCS	$.34_{\pm.04}$	$1.00_{\pm.01}$	$3.8_{\pm.4}$x	$.66_{\pm.04}$	$70_{\pm\ 3}$%

TABLE II: IFT vs. Mettack attack at $k = 1$ edge removal (10 seeds). IFT wins 30/30 comparisons. τ: Kendall correlation with brute-force ground truth.

Dataset	IFT τ	Met τ	IFT dmg	Met dmg
Cora	$+.28_{\pm.07}$	$+.20_{\pm.11}$	$.55_{\pm\ .27}$	$.16_{\pm.09}$
Citeseer	$+.38_{\pm.12}$	$+.16_{\pm.13}$	$1.1\ _{\pm1.5}$	$.11_{\pm.14}$
WikiCS	$+.14_{\pm.04}$	$+.52_{\pm.02}$	$.05_{\pm\ .01}$	$.01_{\pm.00}$

Coverage ranges from 70% (WikiCS) to 92% (Amazon Photo), reflecting nodes with positive classification margin.

B. Attack Comparison: AEGIS vs. Mettack

We compare AEGIS's IFT-based structural attack against Mettack [2] (Meta-Self variant using a 2-layer GCN [34] surrogate). Both methods rank edges for removal; damage is measured by IGNN fixed-point shift $\|\Delta z^*\|$ after reconvergence.

AEGIS inflicts 3–10× more damage than Mettack at every seed and budget level (30/30 wins at $k = 1, \ldots, 5$). However, this comparison has an important caveat: Mettack uses a GCN surrogate, so its poor transfer to IGNN reflects architectural mismatch as much as AEGIS's analytical superiority. Mettack's surrogate occasionally achieves higher brute-force correlation (WikiCS $\tau = +0.52$), but the actual damage is 5× lower because the GCN decision boundary diverges from the IGNN equilibrium. The fair comparison is against the adaptive attacker in section V-D, which uses the same IFT information as AEGIS.

C. Certificate Comparison: Deterministic vs. Smoothing

We compare AEGIS's deterministic first-order radii (theorem 3) against randomized smoothing [9] with Gaussian noise on existing edges ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 Monte Carlo samples, Clopper-Pearson confidence $1 - \alpha = 0.999$). table III reports median certified radius on Cora, Citeseer, and WikiCS (10 seeds, 50-node subgraphs).

The comparison reveals a qualitative difference rather than a simple radius ordering. At $\sigma \leq 0.10$, smoothing radii are *constant* across all nodes and datasets: every node receives the identical maximum-possible radius $r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$ because all 1000 Monte Carlo samples predict correctly. This produces larger absolute radii than AEGIS (0.123 vs. 0.046 at $\sigma = 0.05$) but provides zero per-node differentiation. The certificates do not distinguish between structurally vulnerable

TABLE III: AEGIS deterministic radii vs. smoothing certificates (10 seeds). Smoothing at $\sigma \leq 0.10$ produces identical per-node radii; AEGIS radii vary by node, reflecting structural vulnerability.

Dataset	AEGIS	Smoothing median radius			
		σ=.01	σ=.05	σ=.10	σ=.15
Cora	$.046_{\pm.010}$	.025	.123	$.244_{\pm.005}$	$.283_{\pm.132}$
Citeseer	$.045_{\pm.012}$	.025	.123	$.233_{\pm.013}$	$.091_{\pm.069}$
WikiCS	$.040_{\pm.004}$	.025	.123	$.246_{\pm.000}$	$.341_{\pm.017}$

TABLE IV: Adaptive attack (white-box PGD via IFT gradients on IGNN) vs. AEGIS first-order radii (10 seeds). Breach: fraction of nodes with $\varepsilon < r_v$ whose prediction changes.

Dataset	ε	Breach	AEGIS dmg	Adapt. dmg	Ratio
Cora	0.01	0%	$0.33_{\pm 0.07}$	$0.26_{\pm 0.11}$	0.80
Cora	0.05	0%	$1.72_{\pm 0.35}$	$1.32_{\pm 0.55}$	0.76
Cora	0.10	0%	$3.70_{\pm 0.79}$	$2.69_{\pm 1.18}$	0.72
Citeseer	0.01	0%	$0.40_{\pm 0.08}$	$0.34_{\pm 0.06}$	0.84
Citeseer	0.05	0%	$2.14_{\pm 0.42}$	$1.68_{\pm 0.28}$	0.79
Citeseer	0.10	0%	$4.63_{\pm 0.95}$	$3.37_{\pm 0.55}$	0.74
WikiCS	0.01	0%	$0.21_{\pm 0.02}$	$0.10_{\pm 0.10}$	0.47
WikiCS	0.05	0.3%	$1.04_{\pm 0.11}$	$0.48_{\pm 0.49}$	0.46
WikiCS	0.10	0.6%	$2.10_{\pm 0.22}$	$0.96_{\pm 0.97}$	0.46

and robust nodes. At $\sigma = 0.15$, smoothing begins to differentiate (some nodes lose prediction consistency), but with high variance.

AEGIS radii, by contrast, are structurally informative: median radius 0.041–0.046 with per-node variation that reflects actual equilibrium sensitivity. Nodes in dense, well-connected regions receive larger radii ($r_v \approx 0.09$), while boundary nodes in sensitive positions get smaller ones ($r_v \approx 0.01$). This per-node differentiation is the key advantage for applications that need vulnerability ranking rather than uniform worst-case bounds. The two approaches are complementary: smoothing provides uniform probabilistic guarantees, while AEGIS provides tight, deterministic, structurally differentiated first-order radii.

D. Adaptive Attack Evaluation

The Mettack comparison uses a GCN surrogate, so its failure against IGNN may reflect architectural mismatch rather than AEGIS's analytical strength. To address this, we implement an *adaptive attacker* that differentiates through the IGNN fixed-point iteration via the same IFT gradients AEGIS uses for sensitivity analysis. This attacker optimizes perturbations directly against the IGNN loss surface using projected gradient descent [35] (PGD, 50 steps, step size $\varepsilon/10$).

The adaptive attacker breaches zero nodes within their first-order radius at $\varepsilon = 0.01$ across all datasets, and fewer than 1% even at $\varepsilon = 0.10$ (table IV). This confirms the empirical validity of the first-order radii in the small-budget regime. AEGIS's SVD-optimal attack consistently inflicts 20–50% more damage than adaptive PGD across all tested budgets (ratio 0.46–0.84). The reason is that the SVD direction is the globally optimal first-order perturbation, while PGD explores a non-convex landscape and can become trapped in local optima. The advantage is largest on WikiCS (ratio 0.46), where the high-dimensional perturbation space makes PGD convergence hardest.

TABLE V: First-order radius stability across subgraph sizes on Cora (10 seeds). Tightness and radius are stable for $N \geq 50$.

N	Tight.	Med. r_v	Cert%	ρ	Time
30	$1.012_{\pm.003}$	$.064_{\pm.02}$	$89_{\pm 8}\%$	$.44_{\pm.04}$	0.5s
50	$1.013_{\pm.003}$	$.046_{\pm.01}$	$82_{\pm 8}\%$	$.45_{\pm.04}$	1.3s
100	$1.019_{\pm.004}$	$.030_{\pm.01}$	$84_{\pm 6}\%$	$.61_{\pm.05}$	10.3s
200	$1.031_{\pm.009}$	$.019_{\pm.01}$	$89_{\pm 3}\%$	$.75_{\pm.08}$	86.3s

E. Phase Transition and Scalability

Phase transition. To validate theorem 1, we scale the adjacency to target $\rho \in \{0.3, 0.5, 0.7, 0.85, 0.9, 0.95, 0.99\}$ and measure the fixed-point shift at $\varepsilon = 0.01$. The shift amplifies by 83× as ρ increases from 0.3 to 0.99, confirming the $1/(1-\kappa)$ amplification factor.

Scalability. On a single NVIDIA RTX 4090 GPU, the full AEGIS pipeline takes 0.5s ($N = 20$), 1.3s ($N = 50$), 2.8s ($N = 100$), and 8.1s ($N = 200$), dominated by the $O(D \cdot N^2)$ Jacobian J_A computation. The practical limit is $N \approx 200$ ($D = 12{,}800$), covering all IEEE test cases and typical ego-network sizes.

F. Subgraph Size Ablation

The default $N = 50$ BFS ego-subgraph is critical to scalability. table V evaluates whether analysis quality depends on subgraph size by varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds, same center node).

Tightness is stable at 1.01–1.02 for $N \in \{30, 50, 100\}$ and degrades mildly to 1.031 ± 0.009 at $N = 200$, where the larger subgraph captures higher spectral radius ($\rho = 0.75$ vs. 0.45). Median radius decreases with N (from 0.064 at $N = 30$ to 0.019 at $N = 200$) because larger subgraphs include peripheral nodes with smaller classification margins. Coverage remains stable at 82–89%. We recommend $N = 50$ as the default: tightness 1.013 ± 0.003 with 66× speedup over $N = 200$ (1.3s vs. 86s).

G. Hyperparameter Sensitivity

We evaluate sensitivity to the two key architectural hyperparameters: hidden dimension d and spectral-norm constraint c (the upper bound on $\|W\|_2$).

Hidden dimension. Varying $d \in \{16, 32, 64, 128\}$ on Cora: tightness is 1.012–1.013 for all values, confirming robustness to architecture width. Accuracy improves from 74.7% ($d = 16$) to 79.1% ($d = 64$). The critical budget and median radius are stable across d: $\varepsilon_{\text{crit}} \in [0.53, 0.56]$, $r_v \in [0.037, 0.051]$.

Spectral norm constraint. Varying $c \in \{0.5, 0.7, 0.9, 0.95\}$ reveals the accuracy-robustness tradeoff predicted by theorem 1. At $c = 0.5$: $\rho = 0.25$, $\varepsilon_{\text{crit}} = 1.50$, median $r_v = 0.090$, giving the largest radii but accuracy drops to 72.1%. At

$c = 0.9$: $\rho = 0.45$, $\varepsilon_{\text{crit}} = 0.61$, $r_v = 0.051$, accuracy peaks at 80.6%. Radii shrink by 1.8× from $c = 0.5$ to $c = 0.9$ as $1/(1-\kappa)$ grows, consistent with the theoretical amplification factor in eq. (7).

H. DEQ Convergence Diagnostics

We report convergence statistics for the IGNN fixed-point iteration across all 9 datasets (10 seeds, relative tolerance $\|Z_{k+1} - Z_k\|_F < 10^{-5} \cdot \max(\|Z_k\|_F, 1)$). Convergence is 100% across all datasets and seeds. Graph benchmarks require 28 ± 7 (Amazon) to 113 ± 39 (Cora) iterations, while IEEE grids converge faster: 11 ± 2 (case14) to 44 ± 13 (case57). The pseudospectral index is $\eta = 1.02$–1.05 on graph benchmarks and $\eta = 1.20$–1.28 on IEEE grids, confirming mild non-normality across all domains. No seed on any dataset failed to converge, consistent with the enforced contractivity $\kappa = \|J_z\|_2 < 1$ via spectral normalization [23] of W.

VI. Case Study: Power Flow Contingency

The experiments above validate AEGIS on standard graph benchmarks. A natural question is whether the vulnerability spectrum captures meaningful structure beyond these benchmarks. We demonstrate on a fundamentally different domain, AC power flow, where the adversarial vulnerability spectrum recovers a classical engineering problem: N-1 contingency analysis. Recent work has applied GNNs to power flow computation [36], [37] and contingency screening [38]; AEGIS approaches the same problem from the vulnerability-analysis perspective.

A. Background: N-1 Contingency

Power grid operators must ensure that no single transmission line failure causes system instability. The standard N-1 contingency assessment removes each line in turn, re-solves the power flow equations, and ranks lines by the severity of the resulting voltage and flow deviations [39]. This brute-force procedure scales as $O(|E|)$ full power flow solves, which is expensive for large grids and infeasible for real-time operation.

We observe that AEGIS's per-edge vulnerability $v_{ij} = \|[S_c]_{:,k}\|_2$ computes an analogous quantity via a single IFT analysis. The mathematical correspondence is approximate (continuous first-order sensitivity vs. discrete full-removal contingency), as summarized in the following table:

ML concept	Power systems concept
Edge weight perturbation δA_{ij}	Topology change (line trip)
Vulnerability spectrum v_{ij}	Contingency severity ranking
Critical budget $\varepsilon_{\text{crit}}$	Stability margin
Contractivity loss $\kappa \to 1$	Approach to voltage collapse

Scope of the analogy. Physical line outages are discrete (full edge removal), whereas AEGIS analyzes continuous edge-weight perturbations. Line impedance is fixed; what changes under contingency is the topology and the resulting power flow redistribution. AEGIS captures this as a first-order sensitivity to adjacency changes, which is analogous to how Power Transfer Distribution Factors (PTDF) and Line Outage Distribution Factors (LODF) [39] linearize the effect of generation shifts and line trips, respectively.

B. Experimental Setup

We train ContractiveGCN-PF, an IGNN variant for AC power flow, on four standard IEEE test cases [20]. Training data is generated via PandaPower's [33] full AC Newton-Raphson solver (2,000 load samples per case, uniformly sampled at 70–130% of nominal load). The model takes binary adjacency from the admittance matrix and 5 bus features: bus type indicators (slack, PV), active and reactive power injections (P, Q), and voltage magnitude $|V|$. Unlike graph benchmarks, power flow grids are small enough (14–118 buses) to analyze without subgraph extraction.

Limitation: sample diversity. The 2,000 samples per case cover uniform load scaling but not seasonal variation, renewable intermittency, or generator outage combinations. The AEGIS analysis itself (IFT-based sensitivity) is exact given any trained model, but the model's fidelity to real operating conditions depends on training data diversity.

C. Results

table VI reports constrained tightness and N-1 ranking correlation (Kendall τ between AEGIS vulnerability spectrum and brute-force contingency ranking) across 10 seeds.

Key observations. First, tightness of approximately 1.00 under constrained perturbations transfers to the power flow domain, confirming that the first-order prediction is not domain-specific. Second, ranking correlation ($\tau = 0.37$–0.67) is moderate, reflecting the gap between continuous first-order sensitivity and discrete full-edge-removal contingency. The operationally relevant metric is precision at top-k: AEGIS achieves P@5 between 0.52 and 0.70, and P@10 between 0.66 and 0.81, meaning it correctly identifies 7–8 of the top 10 most critical lines on larger grids.

Third, we compare against an effective-resistance baseline derived from the graph Laplacian pseudoinverse (a simplified DC-approximation proxy for LODF). On small grids (case14/30), this baseline achieves $\tau \approx 0.37$, comparable to AEGIS. On larger grids (case57/118), the simplified baseline degrades ($\tau \leq 0.04$) while AEGIS maintains $\tau = 0.62$–0.67. We note that this baseline uses unit impedance (binary adjacency), not actual line reactances; a proper LODF implementation with real impedance data would likely perform better but requires domain-specific information unavailable to AEGIS.

Practical positioning. AEGIS is not a replacement for PTDF/LODF screening, which exploits domain-specific DC power flow linearization with real line parameters [39]. Rather, it demonstrates that general-purpose IFT-based vulnerability analysis independently recovers contingency structure without domain-specific inputs. For case118, AEGIS computes the ranking in a single IFT analysis (2.3s) versus 179 full AC solves for brute-force N-1, with P@10 of 0.81.

Limitation: operational readiness. The $\tau = 0.37$–0.67 correlation is insufficient for direct operational use without

TABLE VI: AEGIS on IEEE power flow benchmarks (10 seeds). Tightness ≈ 1.00 under constrained perturbations. P@k: precision of AEGIS top-k vs. brute-force N-1 ground truth. EffRes: effective-resistance baseline (simplified LODF proxy).

				AEGIS	AEGIS Precision		EffRes Baseline	
Case	N	$\lvert E\rvert$	Tight.	τ	P@5	P@10	τ	P@10
case14	14	20	$1.00_{\pm.01}$	$+.42_{\pm.19}$	$.70_{\pm.10}$	$.74_{\pm.12}$	$+.37_{\pm.09}$	$.60_{\pm.06}$
case30	30	41	$1.00_{\pm.01}$	$+.37_{\pm.17}$	$.66_{\pm.13}$	$.68_{\pm.06}$	$+.37_{\pm.05}$	$.40_{\pm.08}$
case57	57	78	$1.02_{\pm.01}$	$+.67_{\pm.09}$	$.52_{\pm.24}$	$.66_{\pm.13}$	$-.14_{\pm.11}$	$.06_{\pm.05}$
case118	118	179	$1.01_{\pm.01}$	$+.62_{\pm.11}$	$.62_{\pm.14}$	$.81_{\pm.10}$	$+.04_{\pm.08}$	$.10_{\pm.00}$

verification. Grid operators require near-perfect recall of critical contingencies. AEGIS is positioned as a fast screening layer, not a standalone contingency tool.

VII. Related Work

AEGIS sits at the intersection of adversarial robustness of GNNs, robustness of implicit networks, and sensitivity analysis. We review each thread and clarify how AEGIS differs.

Adversarial attacks on graph structure. Zügner et al. [1] demonstrated GNN vulnerability to structural perturbation with Nettack; Mettack [2] extended this to global poisoning via meta-gradients through a GCN [34] surrogate. Dai et al. [3] framed attacks as reinforcement learning, Bojchevski and Günnemann [40] proposed global poisoning via spectral shifts, and Xu et al. [4] formulated topology attack as bilevel optimization. Black-box variants include GF-Attack [41] via graph signal processing and influence-function approaches [42]. Wu et al. [43] and Jin et al. [15] survey the landscape. All target explicit GNNs with surrogate gradients or heuristic search. AEGIS instead analyzes the implicit equilibrium directly via the IFT, providing a closed-form optimal attack through SVD of S_c.

Certified robustness for GNNs. Bojchevski and Günnemann [12] pioneered certifiable robustness via convex relaxations for GCN-class models. Zügner and Günnemann [11] extended this with interval bound propagation. Both require finite-depth architectures incompatible with implicit models. Randomized smoothing [9] provides probabilistic certificates for any classifier; Scholten et al. refined this with message-interception [10] and hierarchical [44] smoothing. Jia et al. [45] certified community detection via smoothing, Wang et al. [13] proposed message-passing invariance certificates, and Geisler et al. [46] scaled certified defenses to large graphs. Li et al. [14] proposed AGNNCert with deterministic guarantees under arbitrary perturbations. All target explicit architectures. AEGIS fills the gap for implicit models with deterministic first-order radii that are structurally differentiated per node (section V-C).

GNN defense. Empirical defenses improve robustness without formal guarantees: Robust GCN [5] absorbs noise via Gaussian hidden layers, Pro-GNN [7] jointly learns clean structure, GNNGuard [6] reweights edges during aggregation, PA-GNN [47] transfers robustness across domains, PTD-Net [8] learns denoised topologies, and soft-median aggregation [48] replaces mean pooling. Entezari et al. [49] observed that adversarial perturbations primarily affect high-rank spectral components, motivating low-rank SVD defenses. AEGIS is an analytical tool rather than a defense, but its vulnerability spectrum could inform defense design.

DEQ and implicit network robustness. The DEQ framework [21] introduced fixed-point models; extensions include multiscale DEQs [22], Jacobian regularization for stability [26], phantom gradients [50], and Jacobian-free backpropagation [25]. Robustness analysis focuses on *input* perturbation: monotone operator networks [51] provide built-in Lipschitz bounds, Revay et al. [30] use linear matrix inequalities, Pabbaraju et al. [52] estimate tighter Lipschitz constants, and El Ghaoui et al. [29] analyze well-posedness. All address $\|\partial z^*/\partial x\|$, not *structural* sensitivity $\|\partial z^*/\partial A\|$, which changes the operator itself. AEGIS addresses this distinct perturbation type through S_c.

Implicit GNNs. IGNN [16] enforces unique equilibria via spectral normalization [23]. EIGNN [53] improves efficiency through eigendecomposition. Liu et al. [54] study GNN stability without per-node radii or vulnerability characterization. None analyze adversarial structural vulnerability at the equilibrium as AEGIS does.

Sensitivity analysis. AEGIS draws on classical matrix perturbation theory [31], [27]. In deep learning, Novak et al. [55] connected Jacobian sensitivity to generalization and Raghu et al. [56] analyzed expressivity via trajectory lengths. These study feedforward architectures; AEGIS applies sensitivity analysis to implicit fixed points, where the resolvent $(I-J_z)^{-1}$ amplifies perturbation effects by $1/(1-\kappa)$.

GNNs for power systems. Our case study (section VI) connects to GNN-based power flow solvers [36], [37] and GNN contingency screening [38]. AEGIS contributes by showing that equilibrium sensitivity independently recovers contingency rankings without domain-specific linearization.

Positioning. AEGIS applies to contractive implicit GNNs with operator $F(Z,A)=\sigma(\hat{A}ZW^\top+b)$. Explicit architectures (GAT, GIN, GraphSAGE) lack this structure and are outside scope. Within its scope, AEGIS provides tight, structurally differentiated, first-order vulnerability analysis that no existing method offers.

VIII. Conclusion

We presented AEGIS, a framework that exploits the equilibrium structure of contractive implicit GNNs for adversarial vulnerability analysis. Using the operator-norm contraction constant $\kappa = \|J_z\|_2$ and the implicit function theorem, we formalized vulnerability regimes around a critical budget $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\|W\|_2$ with explicit assumptions (ReLU, spectral-norm constraint, linear head). The central technical contribution, the constrained sensitivity matrix S_c, achieves first-order tightness of 1.00 ± 0.01 at $\varepsilon = 0.01$ across 9 datasets spanning 4 domains. AEGIS's SVD-optimal attack inflicts 20–50% more damage than adaptive white-box PGD. Its per-node robustness radii provide structurally differentiated vulnerability assessments that reflect actual equilibrium sensitivity. The vulnerability spectrum independently recovers power grid N-1 contingency rankings (P@10 between 0.66 and 0.81), bridging adversarial graph analysis with power systems engineering.

Limitations. (1) *Architecture scope:* AEGIS applies to contractive implicit GNNs (IGNN-class [16]) only, not to explicit architectures such as GAT, GIN, or GraphSAGE. (2) *First-order approximation:* radii are locally tight for small ε but conservative at large budgets ($\varepsilon > 0.1$). Without bounding the second-order remainder, they are local sensitivity radii rather than global certificates. Edge additions that change the sparsity pattern break the continuous perturbation assumption entirely. (3) *Threat model:* we perturb normalized adjacency $\hat{A}$ directly; perturbations to raw A that change degree normalization require additional chain-rule terms. (4) *Scalability:* dense Jacobian computation limits subgraph size to $N \approx 200$. (5) *Power flow:* Kendall $\tau = 0.37$–0.67 is insufficient for standalone operational use; AEGIS serves as a screening layer, not a replacement for domain-specific PTDF/LODF analysis [39].

Future work. Sparse Neumann-series solvers for larger subgraphs. Extension to monotone operator networks [51] and neural ODEs on graphs. Second-order remainder bounds to upgrade first-order radii into formal certificates. Discrete perturbation models (edge insertion/deletion) via combinatorial optimization over the constrained sensitivity spectrum.

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